

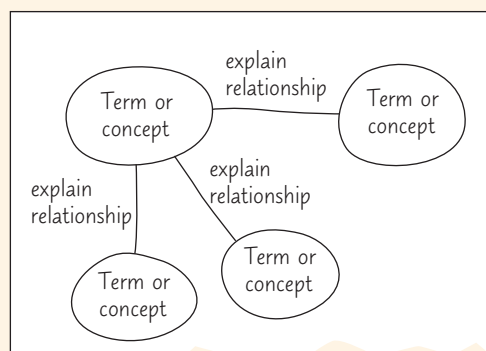
# 22 Descent with Modification: A Darwinian View of Life

## KEY CONCEPTS

- 22.1** The Darwinian revolution challenged traditional views of a young Earth inhabited by unchanging species p. 469
- 22.2** Descent with modification by natural selection explains the adaptations of organisms and the unity and diversity of life p. 471
- 22.3** Evolution is supported by an overwhelming amount of scientific evidence p. 476

## Study Tip

**Make a word cloud:** This chapter covers key topics related to evolution, such as descent with modification, heritable characteristics, natural selection, adaptation, convergent evolution, homology, and unity of life. Draw a “word cloud” of these terms: As you read the chapter, draw a line between terms that affect or relate to one another. Next to that line, briefly explain how those terms are connected.



## ➔ Go to Mastering Biology

**For Students** (in eText and Study Area)

- Get Ready for Chapter 22
- Figure 22.17 Walkthrough: Tree Thinking
- Figure 22.20 Walkthrough: Transition to Life in the Sea
- BBC Video: A Future Without Antibiotics?

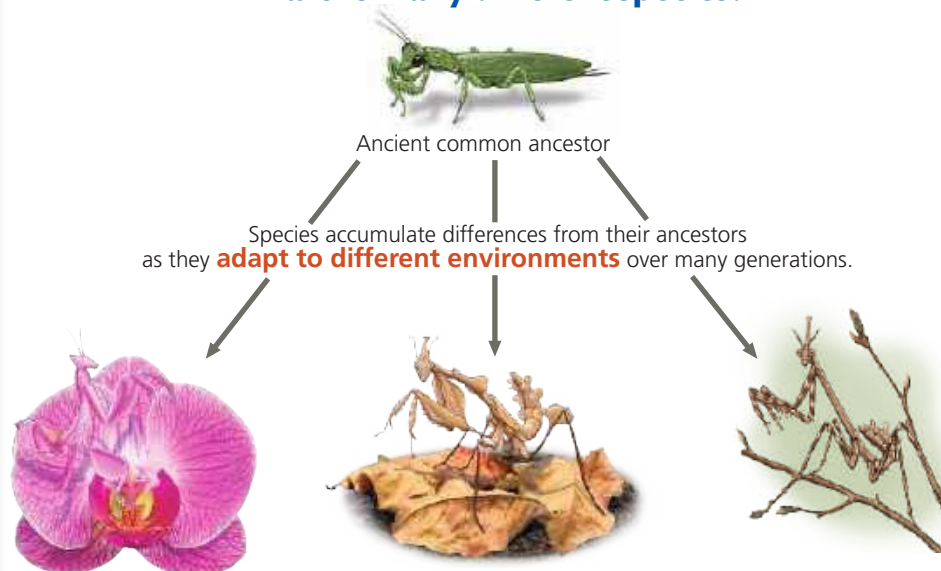
**For Instructors to Assign** (in Item Library)

- HHMI Video: The Making of the Fittest: Natural Selection and Adaptation (Pocket Mouse)



**Figure 22.1** This Malaysian orchid mantis (*Hymenopus coronatus*) closely resembles the flower on which it rests, waiting for unwary prey to come within its reach. Other mantises have diverse shapes and colors that have evolved in different environments—yet all mantises also share certain features, such as grasping forelimbs, large eyes, and six legs.

## What causes the similarities and differences among Earth's many different species?



While different in some ways, these species share many similar features because they descended from a common ancestor. This process of

### descent with modification

shared ancestry,  
resulting in shared  
characteristics

accumulation  
of differences

has given rise to the **diversity of life**.